

Project Plan for Thesis (5 pages)

Title

Enabling Water Reuse on Green Roofs with IoT Monitoring and Control

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Individual Contribution Statement

This thesis is conducted as an individual project. The author is responsible for system design, hardware integration, software development, data collection, analysis, and thesis writing.

Researcher Suggesting This Idea

Akash Gaur

In Collaboration With Company

Not applicable

Keywords

Green roofs; water reuse; IoT monitoring; ESP32; Raspberry Pi; MQTT; sensor integration; control systems

Motivation

Urban green roofs play an important role in sustainable city development by reducing stormwater runoff, improving local microclimate, and supporting biodiversity. Despite these benefits, many green roof installations lack continuous monitoring and rarely reuse captured rainwater for irrigation. Existing systems are often proprietary, costly, and difficult to adapt or maintain.

Recent advances in low-cost IoT hardware and open-source platforms enable scalable monitoring and basic control of environmental systems. By integrating sensors, edge devices, and cloud-based visualization, it is possible to create a closed-loop system that supports water reuse in a reliable and maintainable way. This project is motivated by the need for practical, reproducible solutions that demonstrate how IoT technologies can improve water efficiency and operational insight in green roof installations.

Contribution

This work contributes to contemporary research by demonstrating a complete, open-source IoT architecture for monitoring and controlling water reuse on green roofs. Unlike many conceptual or simulation-based studies, the project focuses on a real-world prototype with battery-powered edge nodes, gateway-based control logic, and cloud integration.

The contribution lies in:

- A systems-engineering implementation combining sensing, communication, and actuation.
 - Evaluation of data reliability, control safety, and water reuse effectiveness.
 - Practical deployment insights that can be reused in future academic or municipal projects.
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Research Question(s)

How can water reuse on green roofs be enabled using IoT-based monitoring and control systems?

Methodology

This study adopts a **quantitative, experimental research approach**. A functional IoT system is designed, implemented, and evaluated through controlled experiments and real-world operation.

Research Strategy

The research follows a design-and-evaluate strategy:

1. Design an IoT architecture for green roof monitoring and water reuse.
2. Implement hardware, firmware, gateway, and cloud components.
3. Collect quantitative sensor and system performance data.
4. Evaluate the system using predefined metrics.

Methods Discussion

Quantitative data are collected from sensors measuring environmental and system variables. System performance is analyzed using descriptive statistics and event-based comparisons between different operating scenarios (e.g., control enabled vs. disabled).

Variables

Dependent Variables

- Soil moisture levels
- Volume of reused water for irrigation
- Data reliability (packet loss, missing values)
- Control safety indicators (overflow, dry-run events)
- Battery endurance (operational time)

Independent Variables

- System control mode (automatic control vs. monitoring only)
 - Environmental conditions (rainfall, dry periods)
 - Sensor placement and sampling intervals
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Data Set / Data / Code Base

Data

The dataset consists of time-stamped sensor data collected from the deployed system, including:

- Soil moisture (two depths)
- Soil temperature
- Rainfall
- Drain level or flow
- Air temperature and humidity
- Actuator states (pump and valve)
- Battery voltage

Data are stored locally on the gateway and forwarded to a cloud platform for visualization and export.

Data Creation

All data are generated by the system during experiments. No external or proprietary datasets are used.

Code Base

All software is developed by the author using open-source tools:

- ESP32 firmware (Arduino framework)
- MQTT communication (Mosquitto)
- Gateway services (Raspberry Pi, optional Node-RED)
- Cloud dashboard (ThingsBoard or OpenRemote)

Experiment Design / Implementation

Computer Specification

- Raspberry Pi (gateway)
- ESP32 microcontrollers (edge nodes)
- Linux-based development environment

Architecture

- **Edge Layer:** ESP32 nodes with sensors and battery power
- **Gateway Layer:** Raspberry Pi handling MQTT, buffering, and control logic
- **Cloud Layer:** Visualization and rule-based monitoring

Hardware Setup

- Soil moisture and temperature sensors
- Rain gauge
- Flow or level sensor
- Solenoid valve and DC pump
- Battery power supply

Control Logic

Threshold-based irrigation logic is implemented with safety constraints, including maximum runtime and dry-run protection.

Time Plan

Week	Task	Planned Time
1	Literature review, requirements analysis, project planning	40 h
2	Initial hardware setup and basic sensor integration	40 h
3	MQTT communication and gateway setup	40 h
4	Cloud integration and dashboard design	40 h
5–6	Full sensor integration and battery optimization	80 h
7–8	Actuator integration and control logic	80 h
9–10	Field experiments and data collection	80 h
11–12	Data analysis and evaluation	80 h
13–15	Writing and revision	120 h
16	Final submission preparation	40 h

Potential Results

It is expected that the implemented system will demonstrate reliable data collection and enable partial reuse of harvested rainwater for irrigation. The results are expected to show that IoT-based monitoring and control can improve water efficiency while maintaining safe operation.

The findings are expected to answer the research question by demonstrating a feasible technical solution. While results are context-specific, the architecture and methodology can be generalized to similar green roof or small-scale water reuse applications.

References

References will be managed using a reference handling system and formatted according to the required academic standard.

- MQTT Specification
- ThingsBoard Documentation
- OpenRemote Documentation
- Espressif ESP32 Technical Reference Manual
- Selected peer-reviewed literature on green roofs and stormwater management